

Rethinking r-PKP: a New Formulation for the Relaxed Permuted Kernel Problem

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Abstract. Among the schemes in the second round of NIST’s additional call for Post-Quantum signatures, PERK builds its security on the intractability of the Permuted Kernel Problem (PKP). In its original formulation, this problem asks, on input three matrices $\mathbf{H}, \mathbf{X}, \mathbf{Y}$, to find a permutation matrix $\mathbf{P}$ such that $\mathbf{HPX} = \mathbf{Y}$. To achieve better performance and smaller signatures, in its first proposal, the PERK signature modified the security assumption in the following way: given a PKP instance, the matrix $\mathbf{P}$ does not have to verify the exact previous equation but a relaxed one, taking care of a non-null vector $\mathbf{v}$ such that $(\mathbf{HPX})\mathbf{v} = \mathbf{Y}\mathbf{v}$.

In this work, we rephrase the relaxed problem so that it no longer depends on the PKP instance nor the vector $\mathbf{v}$. We show that it suffices to find $\mathbf{P}$ such that $\mathbf{HPX} - \mathbf{Y}$ has rank deficiency. This generalized formulation is easier to model and allows us to design an algebraic attack inspired by those of MinRank and Rank Syndrome Decoding, writing a polynomial system in the entries of $\mathbf{P}$. Moreover, we can consider it as linear in the minors of $\mathbf{P}$ and provide some results on them, which may be of independent interest.

Keywords: Post-Quantum Cryptography, Permuted Kernel Problem, Algebraic Attacks

1 Introduction

With NIST’s call currently in the second round of the standardization process for new post-quantum digital signatures, increasing attention is being paid to the cryptanalysis of the fourteen proposals remaining in the competition. One of these proposals is PERK [1,9], which bases its security on the Permuted Kernel Problem (PKP). Informally, we are given an instance of matrices $(\mathbf{H}, \mathbf{X}) \in \mathbb{F}_q^{m \times n} \times \mathbb{F}_q^{n \times t}$ and it is required to find a permutation matrix $\mathbf{P}$ such that $\mathbf{HPX} = \mathbf{0}$. This problem has been proven to be NP-hard [41] and has received a lot of attention from the cryptographic community in recent years. The best attacks currently

available are combinatorial, based on Information Set Decoding (ISD) [40,15], and, to the best of our knowledge, no algebraic attacks which outperform ISD have ever been proposed.

The presumed hardness of the problem has led to two variants of the original PKP. The first one is an inhomogeneous version of PKP, which we denote IPKP, where we are given a triple of matrices $(\mathbf{H}, \mathbf{X}, \mathbf{Y}) \in \mathbb{F}_q^{m \times n} \times \mathbb{F}_q^{n \times t} \times \mathbb{F}_q^{m \times t}$ and we aim to find a permutation matrix $\mathbf{P}$ such that $\mathbf{H}\mathbf{P}\mathbf{X} = \mathbf{Y}$. The second variant is a *relaxed* version of PKP, where we want to find a permutation matrix $\mathbf{P}$ for which the above equation holds when it is multiplied by a non-null vector v , namely $(\mathbf{H}\mathbf{P}\mathbf{X})v = \mathbf{Y}v$. The latter was introduced in 2023 for the first proposal of the PERK signature, albeit not independently from the original problem: to describe it, authors of [9] start from a non-relaxed PKP instance. We wonder which impact can have on its definition if we drop this requirement and give the problem a standalone description. In 2025, PERK’s team announced that the new version of PERK will rely directly on PKP.

1.1 Attacks to PKP and IPKP

The Permuted Kernel Problem and its inhomogeneous variant have been extensively studied in recent years. It is well-known that PKP is a NP-hard problem [41]. Furthermore, in [40] it has been shown that IPKP is equivalent to the Permutation Subcode Equivalence problem [8].

The first attack to PKP was presented by Georgiades in [29], exploiting the use of symmetric polynomials to reduce the number of suitable permutations. It was followed by the work of Baritaud, Campana, Chauvaud and Gilbert [6], which trades some memory in order to achieve better performance. Both these papers break the parameters set suggested by Shamir for its identification scheme [41]. Later, Patarin and Chauvaud [37] improve the state of the art of the attacks to PKP with a new time-memory trade-off. Jaulmes and Joux [35] present a new method for solving PKP: they adapt an algorithm for counting points in an elliptic curve to solve an intermediate problem to which PKP can be reduced. For a long time this has been believed to be the state of the art for cryptographic parameters, but this was disproved in the analysis done by Koussa, Macario-Rat and Patarin in [36]. In the same work [36], the authors propose a novel algorithm using a meet-in-the-middle strategy. This strategy has been then improved by Santini, Baldi, and Chiaraluce [40] and, eventually, by Budroni, Defranceschi, and Pintore [15]. Recently, the expected number of solutions to a random instance of PKP in terms of the parameters has been computed [39]. In [9], the authors observe that for $t \geq n$ we obtain a polynomial time attack. Hence, generally, we want a small t to ensure the hardness of the problem.

1.2 Attacks to r-PKP and r-IPKP

There is not much literature regarding the cryptanalysis of r-IPKP, and the few attacks proposed have only a combinatorial nature. The approach followed in [9] to attack it is the following. One can observe that an instance $(\mathbf{H}, \mathbf{X}, \mathbf{Y})$

of r-IPKP with parameters (q, n, m, t) can be reduced to IPKP with parameters $(q, n, m, 1)$ taking any linear combinations of the columns of both $\mathbf{X}$ and $\mathbf{Y}$. The authors look for a low-weight linear combination of such columns via the Information Set Decoding algorithm, and then the obtained IPKP instance is solved.

1.3 Algebraic attacks to MinRank and Rank Syndrome Decoding

Cryptographic primitives can be analyzed from both a combinatorial and an algebraic point of view. RYDE and Mirath [12, 2], two other schemes currently in the second round of NIST's call, can also be attacked using algebraic techniques. In particular, these modelings build linear equations using as unknowns the *minors* of some matrices. In the following, we describe the Support Minors algorithm [4] for the MinRank problem. A similar algorithm for the Rank Syndrome Decoding problem is described in [3].

In the MinRank problem, the instance is given by matrices $(\mathbf{M}_0, \dots, \mathbf{M}_k)$ of dimension $m \times n$, and the goal is to find a vector of coefficients $\alpha = (\alpha_1, \dots, \alpha_k) \in \mathbb{F}_q^k$ such that the matrix $\mathbf{M} = \mathbf{M}_0 + \sum_{i=1}^k \alpha_i \mathbf{M}_i$ has rank equal to r . The starting point of the Support Minors algorithm is to represent $\mathbf{M}$ as product of matrices $\mathbf{S}\mathbf{C}$, with $\mathbf{S} \in \mathbb{F}_q^{m \times r}$ and $\mathbf{C} \in \mathbb{F}_q^{r \times n}$. Let $\mathbf{m}_j$ be the j -th row of $\mathbf{M}$: by definition, this row belongs to the rowspace of $\mathbf{C}$. It follows that, for each j , we can build the new matrix $\mathbf{C}_j = \begin{bmatrix} \mathbf{m}_j \\ \mathbf{C} \end{bmatrix}$, which has dimension $(r+1) \times n$ and rank at most r . It is then a well known linear algebra fact that all the maximal minors of $\mathbf{C}_j$, that is, the determinants of all possible submatrices of dimension $(r+1) \times (r+1)$ of $\mathbf{C}_j$, are equal to 0. If we expand the determinant with respect to the first row $\mathbf{m}_j$, we get a *bilinear system* where the unknowns are $(\alpha_1, \dots, \alpha_k)$ and the minors of $\mathbf{C}$. We end up with a system with $m \cdot \binom{n}{r+1}$ bilinear equations on $\mathbb{F}_q$ and $k + \binom{n}{r}$ unknowns, that can be solved in different ways, depending on the parameters.

1.4 Our Contribution

We present a new formulation of the relaxed Permuted Kernel Problem. The original statement asks, given matrices $\mathbf{H}, \mathbf{X}, \mathbf{Y}$ which satisfy the PKP equation, to find a permutation matrix $\mathbf{P}$ and a non-null vector $\mathbf{v}$ such that $(\mathbf{H}\mathbf{P}\mathbf{X})\mathbf{v} = \mathbf{Y}\mathbf{v}$. First, we restate this fact observing that we can get rid of the vector $\mathbf{v}$ and searching for a permutation matrix $\mathbf{P}$ such that $\mathbf{M}_\mathbf{P} = \mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}$ is not full rank. Then, we generalize the problem without mentioning the PKP constraint, such that starting from any instance $\mathbf{H}, \mathbf{X}, \mathbf{Y}$ we look for a permutation matrix $\mathbf{P}$ such that $\mathbf{M}_\mathbf{P} = \mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}$ is not full rank. We call this problem generalized Relaxed Permuted Kernel Problem (gr-IPKP). This is a more general and more streamlined formulation: we only deal with the matrix $\mathbf{P}$, and finding it implies solving PERK's definition of r-PKP.

As a consequence, a cleaner formulation is easier to cryptanalyze: we use this new definition to propose the first algebraic attacks for r-PKP and r-IPKP.

We highlight how these are the first attacks targeting r -IPKP as a *standalone* problem, i.e. gr -IPKP. Indeed, in [9], an ad-hoc attack for the PERK scheme is presented, since the r -IPKP instance is an a priori IPKP instance. In this paper, we do not assume this fact. Similarly to what was done for MinRank and Rank Syndrome Decoding, the algebraic systems model the fact that every maximal minor of $\mathbf{M}_P$, i.e. each entry of the compound matrix of maximal order $C(\mathbf{M}_P)$, is zero, and we obtain a set of linear equations where the unknowns are the minors of the secret permutation matrix P . Despite the fact that these models are still not practical, especially with the parameters used by recently proposed schemes that rely on PKP and its variant, we believe they can represent a starting point towards the first practical algebraic attacks against this assumption. In Table 1 we recap the different models we obtain starting by the fact that $\mathbf{M}_P$ is not full rank, highlighting the number of equations, variables, monomials (for linearization) and the degree of the system. It should be noted that it is always possible to add equations defining permutation matrices or the compound matrices of a permutation to the polynomial system; since they do not depend on the modeling, we do not include them in the computation. We estimate the cost of solving gr -IPKP via Gröbner bases, employing the Castelnuovo-Mumford regularity [17,18] which depends on the shape of the ideals involved, and not on the modeling. Such estimations partially derives from the analysis given in [16] and do not change if the instance is homogeneous or not. For parameters n, m, t, q , we provide that the number of bit operations to find a solution of the instance is given by

$$\mathcal{O}\left(\log_2(q)\binom{n^2 + t + n - 2}{t + n - 2}^\omega\right),$$

where $\omega \in [2, 3]$ is the linear algebra constant. A key result in this analysis is the computation of the Castelnuovo-Mumford regularity of the ideal of permutation matrices. We show that for $n \times n$ permutation matrices, in any field, the regularity is equal to $n - 1$. This can be of independent interest for algebraic models concerning such objects, like the ones for the Linear Code Equivalence problem [38,7].

Additionally, we present some results on the minors of a permutation matrix P , which form the compound matrix of P . We show that it is a signed permutation matrix and its structure can be exploited to add more equations.

The paper is divided as follows: Section 2 introduces the notation, defines the assumptions, and recaps some linear algebra tools which will be necessary for the new modelings, while Section 3 reformulates and then generalizes the relaxed Permuted Kernel Problem, showing the links between these new definitions and the original one. Section 4 introduces the new algebraic modelings, both in the inhomogeneous and in the homogeneous case, and the complexity of solving them is computed via the solving degree and the Castelnuovo-Mumford regularity of the ideals involved. Then, Section 5 recaps some results on the behavior of the minors of a permutation matrix, which may also be of independent interest and exploit them to add more equations to the modelings from the previous sections.

	Modeling	Eq.	Var.	Mon.	Deg.
gr-IPKP	Th. 12	$\binom{m}{t}$	n^2	$\sum_{i=1}^t \binom{n}{i}^2$	up to t
	Prop. 10 + Th. 13	$\binom{m-t}{t}$	n^2	$\binom{n}{t}^2$	t
	Subs. 5.1	$\binom{m}{t} + \sum_{i=1}^t (2\binom{n}{i} - 1)$	n^2	$\sum_{i=1}^t \binom{n}{i}^2 i!$	up to t
	Prop. 10 + Subs. 5.1	$\binom{m-t}{t} + 2\binom{n}{t} - 1$	n^2	$\binom{n}{t}^2 t!$	t
gr-PKP	Th. 13	$\binom{m}{t}$	n^2	$\binom{n}{t}^2$	t
	Subs. 5.1	$\binom{m}{t} + 2\binom{n}{t} - 1$	n^2	$\binom{n}{t}^2 t!$	t

Table 1: For each proposed modeling for the inequality $\text{rk}(\mathbf{M}_P) < t$, the first column states if the starting generalized PKP instance is inhomogeneous or not, **Eq.**, **Var.**, **Mon.** count the number of equations, variables and monomials, respectively, and **Deg.** is the degree of the polynomial system in the entries of $\mathbf{P}$.

2 Preliminaries

2.1 Notation

Let $\mathbb{F}_q$ be the finite field with q elements. The cardinality of a finite set S is denoted with $|S|$. The vector space of $n \times m$ matrices with coefficients in $\mathbb{F}_q$ is denoted with $\mathbb{F}_q^{n \times m}$. Given $\mathbf{A} \in \mathbb{F}_q^{n \times m}$ and two sets of indices $I \subset \{1, \dots, n\}, J \subset \{1, \dots, m\}$, we denote with $\mathbf{A}_{I,J}$ the $|I| \times |J|$ submatrix $(\mathbf{A}_{i,j})_{i \in I, j \in J}$. With $\mathbf{A}_{*,J}$ we mean $I = \{1, \dots, n\}$ and analogously for $\mathbf{A}_{I,*}$. The $n \times n$ identity matrix is denoted with $\mathbf{I}_n$. Given a matrix $\mathbf{A} \in \mathbb{F}_q^{n \times m}$, we denote with $\text{Im}(\mathbf{A}) = \{\mathbf{A}v \in \mathbb{F}_q^m \mid v \in \mathbb{F}_q^{n \times 1}\}$ and $\text{RKer}(\mathbf{A}) = \{v \in \mathbb{F}_q^{n \times 1} \mid \mathbf{A}v = 0\}$ the image and right kernel of the linear map associated to $\mathbf{A}$, respectively. With $[n]$ we denote the set of integers $\{1, \dots, n\}$ and with $\binom{[n]}{k}$ the collection of subsets of $[n]$ of cardinality k . A permutation matrix is a square matrix having exactly a single 1 in each row and column and 0 otherwise. A signed permutation matrix is a permutation matrix having ± 1 in its non-zero entries.

2.2 Permuted Kernel Problem and variants

The Permuted Kernel Problem is a computational problem firstly introduced in cryptographic literature by Shamir in [41]. We here present the inhomogeneous version, which is its natural generalization.

Definition 1 (Permuted Kernel Problem). *Let m, n, t be positive integers and q be a prime power. The Inhomogeneous Permuted Kernel Problem (IPKP)*

asks on input $\mathbf{H} \in \mathbb{F}_q^{m \times n}$, $\mathbf{X} \in \mathbb{F}_q^{n \times t}$ and $\mathbf{Y} \in \mathbb{F}_q^{m \times t}$ to find a permutation matrix $\mathbf{P} \in \mathbb{F}_q^{n \times n}$ such that

$$\mathbf{H}\mathbf{P}\mathbf{X} = \mathbf{Y}. \quad (1)$$

Whenever $\mathbf{Y} = \mathbf{0}$, it is known as Permuted Kernel Problem (PKP).

Remark 2. As noted in [1], it is possible to reduce the inhomogeneous case to the homogeneous one by replacing m with $m - t$. Let $\mathbf{H}, \mathbf{X}, \mathbf{Y}$ be an instance of IPKP, we then enlarge the code whose parity check matrix is given by $\mathbf{H}$ with codewords $\mathbf{c}_i$ such that $\mathbf{H}\mathbf{c}_i = \mathbf{y}_i$, where $\mathbf{y}_i$ is the i -th column of $\mathbf{Y}$. If we set $\tilde{\mathbf{H}}$ as the parity check matrix of this enlarged code, the PKP instance is given by $(\tilde{\mathbf{H}}, \mathbf{X})$.

For $t = 1$, the problem is also known as the *mono-dimensional* IPKP. Several schemes from the literature base their security on the presumed (post-quantum) hardness of PKP and IPKP [1,9,10,11,13]. Among these, PERK [1,9] is a recently announced round 2 candidate in the ongoing NIST’s call for post-quantum digital signatures³.

It consists of a Fiat-Shamir signature based on the MPC in the Head (MPCitH) framework [34], proving the knowledge of a *relaxed* variant of the IPKP problem. In particular, PERK’s key generation returns an IPKP instance and one solution as public/secret key pair. The soundness of the zero-knowledge protocol obtained in the MPCitH framework is then reduced to the following problem.

Definition 3 (Relaxed Permuted Kernel Problem). *Let m, n, t be positive integers and q be a prime power, let $\mathbf{H} \in \mathbb{F}_q^{m \times n}$, $\mathbf{X} \in \mathbb{F}_q^{n \times t}$ and $\mathbf{Y} \in \mathbb{F}_q^{m \times t}$ matrices for which there exists a permutation matrix $\mathbf{P}$ such that Equation (1) is satisfied. The relaxed Inhomogeneous Permuted Kernel Problem (r-IPKP) asks on input $\mathbf{H}, \mathbf{X}$, and $\mathbf{Y}$ to find a permutation matrix $\tilde{\mathbf{P}} \in \mathbb{F}_q^{n \times n}$ and a vector $\mathbf{v} \in \mathbb{F}_q^t \setminus \{\mathbf{0}\}$ such that*

$$(\mathbf{H}\tilde{\mathbf{P}}\mathbf{X})\mathbf{v} = \mathbf{Y}\mathbf{v}.$$

Whenever $\mathbf{Y} = \mathbf{0}$, it is called relaxed Permuted Kernel Problem (r-PKP).

It is important to notice that, in the above formulation, r-IPKP is not a standalone assumption: the given inputs already satisfy an IPKP instance and cannot be chosen independently.

Moreover, in defining this problem, it is not clear if authors of [9] require vector $\mathbf{v}$ to be among the outputs; Definition 3 humbly attempts to clarify the issue. In any case, this supports the urge for a new formulation, allowing for a more streamlined description.

Unless otherwise stated, we assume in the following that $n \geq m \geq t$, as for the parameter sets of PERK [1].

³ <https://csrc.nist.gov/projects/pqc-dig-sig/round-2-additional-signatures>, accessed on 26/03/2026.

2.3 Cauchy-Binet Formula

In the second part of the paper we will deal with determinants of products of matrices. For this reason, we report here the classical result known as Cauchy-Binet formula. Let $\mathbf{A} \in \mathbb{F}_q^{\ell \times k}$, $\mathbf{B} \in \mathbb{F}_q^{k \times \ell}$, then

$$\det(\mathbf{AB}) = \sum_{T \in \binom{[k]}{\ell}} \det(\mathbf{A}_{*,T}) \det(\mathbf{B}_{T,*}).$$

For convenience, we report the generalisation to the matrices' minors of the above formula from [31, Prop. 7.182].

Lemma 4. *Let $\mathbf{A} \in \mathbb{F}_q^{\ell \times k}$, $\mathbf{B} \in \mathbb{F}_q^{k \times m}$ and $I \in \binom{[\ell]}{u}$, $J \in \binom{[m]}{u}$ for some $1 \leq u \leq \min\{l, m\}$, then*

$$\det((\mathbf{AB})_{I,J}) = \sum_{T \in \binom{[k]}{u}} \det(\mathbf{A}_{I,T}) \det(\mathbf{B}_{T,J}).$$

2.4 Compound Matrices

We report the definition of the t -th compound matrix (see, for example, [28]). In the rest of the paper, we assume the standard lexicographic order on subsets $I \in \binom{[n]}{t}$; therefore, there is an implicit bijection between the integers $1, \dots, \binom{n}{t}$ and the sets in $\binom{[n]}{t}$.

Definition 5. *Let $\mathbf{A} \in \mathbb{F}_q^{n \times m}$ and let $t \leq \min\{n, m\}$. The t -th compound matrix of $\mathbf{A}$ is the matrix*

$$C_t(\mathbf{A}) = (\det(\mathbf{A}_{I,J}))_{I,J},$$

whose rows are indexed by $I \in \binom{[n]}{t}$ and columns by $J \in \binom{[m]}{t}$.

2.5 Solving a Polynomial System

Given a system of s polynomials $f_1, \dots, f_s \in \mathbb{F}_q[x_1, \dots, x_k]$, a way to find a common root is to compute a Gröbner bases of the ideal they generate. We estimate the complexity of standard linear algebra-based algorithms like F4/F5 [26,23]. Knowledge of a lexicographic Gröbner basis for the ideal generated by $\{f_1, \dots, f_s\}$ provides a solution of the system in polynomial time, using the so-called Shape Lemma (see, for example, [30]).

Algorithms using linear algebra rely on manipulations of the Macaulay matrix of increasing degree. The maximal degree attained by such algorithms is called *solving degree* of the set $\mathcal{F} = \{f_1, \dots, f_s\}$, denoted with $d_{\text{sol}}(\mathcal{F})$. Therefore, a good estimate of the solving degree of a polynomial system provides an upper bound to the complexity of finding the solution of the system. The Macaulay matrix at degree d has rows indexed by multiples of the polynomials $f_1, \dots, f_s$ and columns indexed by monomials of degree less than d . Hence, since the number

of monomials of bounded degree is $\binom{k+d}{d}$, the binary operations needed to solve a system in $\mathbb{F}_q$, using Gauss reductions, is given by

$$\mathcal{O} \left(\log_2(q) \binom{k + d_{\text{sol}}(\mathcal{F})}{d_{\text{sol}}(\mathcal{F})}^\omega \right),$$

where $2 \leq \omega < 3$ is the linear algebra constant.

Estimating the solving degree of the generators of an ideal is the crucial part of the complexity analysis of solving a polynomial system. Basically, it is a measure of the distance of the set $\{f_1, \dots, f_n\}$ from being a Gröbner basis for the ideal $I = (f_1, \dots, f_n)$. In particular, finding bounds on the solving degree is highly non-trivial in general. However, in literature, this estimation is aided by other metrics like the last fall degree [33,32], the degree of regularity [5] or the Castelnuovo-Mumford regularity [17,18]. The solving degree can be well estimated using the degree of regularity only for “sufficiently random” polynomial systems, for example when the polynomials form a *semi-regular* sequence [14,27]. This is the approach followed in [24,22] for estimating the solving degree for polynomial systems arising from HFE and Oil and Vinegar schemes.

3 A Novel Formulation of r-IPKP

Recall that, in the r-IPKP problem from [9], we are given an IPKP instance $(\mathbf{H}, \mathbf{X}, \mathbf{Y})$ and we aim to find a permutation matrix $\mathbf{P}$ and a vector $\mathbf{v}$ such that $(\mathbf{H}\mathbf{P}\mathbf{X})\mathbf{v} = \mathbf{Y}\mathbf{v}$. We can reformulate the problem as the task of finding a permutation matrix $\mathbf{P}$ such that the rank of $\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}$ is not full. In fact, one can find a non-zero vector $\mathbf{v}$ in the right kernel of $\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}$. We can state this fact formally.

Definition 6 (r-IPKP, rephrased). *Let m, n, t be positive integers and q be a prime power, let $(\mathbf{H}, \mathbf{X}, \mathbf{Y}) \in \mathbb{F}_q^{m \times n} \times \mathbb{F}_q^{n \times t} \times \mathbb{F}_q^{m \times t}$ be an IPKP instance. Then, the relaxed Inhomogeneous Permuted Kernel Problem asks on input $\mathbf{H}, \mathbf{X}$ and $\mathbf{Y}$ to find a permutation matrix $\mathbf{P} \in \mathbb{F}_q^{n \times n}$ such that*

$$\text{rk}(\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}) < \min\{m, t\}.$$

Whenever $\mathbf{Y} = \mathbf{0}$, it is called relaxed Permuted Kernel Problem (r-PKP).

We show that this formulation is equivalent to the one given in Definition 3.

Proposition 7. *Every instance of r-IPKP from Definition 3 satisfies Definition 6, inducing the same permutation matrix as solution, and viceversa. The same holds for the homogeneous case.*

Proof. Let $(\mathbf{H}, \mathbf{X}, \mathbf{Y})$ be a r-IPKP instance, and suppose that Definition 3 holds, i.e., there exists a permutation matrix $\mathbf{P}$ and a non-null vector $\mathbf{v}$ such that

$(\mathbf{H}\mathbf{P}\mathbf{X})\mathbf{v} = \mathbf{Y}\mathbf{v}$. This implies that the matrix $(\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}) \in \mathbb{F}_q^{m \times t}$ has a non-null vector in its right kernel. Using the rank-nullity theorem, we can conclude that the matrix $(\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y})$ cannot have maximum rank, that is

$$\text{rk}(\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}) < \min\{m, t\}.$$

The reverse implication follows by reversing the above argument. The same reasoning can be used in the r-PKP case, by simply taking an r-IPKP instance with $\mathbf{Y} = \mathbf{0}$. $\square$

Thanks to the reformulation in Definition 6, we drop the dependence on vector $\mathbf{v}$; as a consequence, there is no reason anymore to restrict ourselves to an IPKP instance: we can remove this constraint and give as input a generic instance $(\mathbf{H}, \mathbf{X}, \mathbf{Y}) \in \mathbb{F}_q^{m \times n} \times \mathbb{F}_q^{n \times t} \times \mathbb{F}_q^{m \times t}$.

Definition 8 (r-IPKP, generalized). *Let m, n, t be positive integers and q be a prime power, let $(\mathbf{H}, \mathbf{X}, \mathbf{Y}) \in \mathbb{F}_q^{m \times n} \times \mathbb{F}_q^{n \times t} \times \mathbb{F}_q^{m \times t}$. The generalized relaxed Inhomogeneous Permuted Kernel Problem (gr-IPKP) asks on input $\mathbf{H}, \mathbf{X}$ and $\mathbf{Y}$ to find a permutation matrix $\mathbf{P} \in \mathbb{F}_q^{n \times n}$ such that*

$$\text{rk}(\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}) < \min\{m, t\}.$$

Whenever $\mathbf{Y} = \mathbf{0}$, we refer to it as generalized relaxed Permuted Kernel Problem (gr-PKP).

We thus obtain an independent assumption with a cleaner formulation, easier to cryptanalyze, and to which the original r-IPKP reduces: indeed, solving our problem implies solving PERK's formulation, as the following proposition shows.

Proposition 9. *The problem r-IPKP (Definition 6) reduces to gr-IPKP (Definition 8). The same holds for the homogeneous case.*

Proof. Let $(\mathbf{H}, \mathbf{X}, \mathbf{Y})$ be a r-IPKP instance; since there is a permutation matrix $\mathbf{P}$ such that $\mathbf{H}\mathbf{P}\mathbf{X} = \mathbf{Y}$, the rank of $\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}$ is zero. Hence, solving $(\mathbf{H}, \mathbf{X}, \mathbf{Y})$ as a gr-IPKP instance gives a solution $\tilde{\mathbf{P}}$ such that the rank of $\mathbf{H}\tilde{\mathbf{P}}\mathbf{X} - \mathbf{Y}$ is non-maximal. Hence, the matrix $\tilde{\mathbf{P}}$ is a solution of the r-IPKP instance too.

The same argument can be used in the r-PKP case, by simply taking an r-IPKP instance with $\mathbf{Y} = \mathbf{0}$. $\square$

The viceversa, however, does not hold. For example, the attack from [9] does not apply to a gr-IPKP instance $(\mathbf{H}, \mathbf{X}, \mathbf{Y})$ for which there does not exist a permutation matrix $\mathbf{P}$ such that $\mathbf{H}\mathbf{P}\mathbf{X} = \mathbf{Y}$.

Reducing to the homogeneous case. Due to Proposition 7, one can reduce a generalized inhomogeneous relaxed PKP instance to a homogeneous one using the same technique introduced in Remark 2 (Section 2.2). This leads to the following result.

Proposition 10. *Let $\mathbf{H} \in \mathbb{F}_q^{m \times n}$, $\mathbf{X} \in \mathbb{F}_q^{n \times t}$, and $\mathbf{Y} \in \mathbb{F}_q^{m \times t}$ be a randomly generated gr-IPKP instance. If $2t < m$, for an overwhelming fraction of gr-IPKP instances with parameters (q, n, m, t) there exists a reduction to gr-PKP with parameters $(q, n, m - t, t)$.*

Proof. Consider $\mathbf{Y}^T$ as the parity-check matrix of a $(m - t)$ -dimensional linear code of length m , having generator matrix $\mathbf{G} \in \mathbb{F}_q^{(m-t) \times m}$. Then, set $\tilde{\mathbf{H}} = \mathbf{G}\mathbf{H}$ and consider the gr-PKP instance $(\tilde{\mathbf{H}}, \mathbf{X})$. Let $\mathbf{P}$ be a permutation matrix such that

$$\text{rk}(\tilde{\mathbf{H}}\mathbf{P}\mathbf{X}) < t.$$

This reduction works whenever, for the above $\mathbf{P}$, we have

$$\text{rk}(\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}) < t,$$

and this happens if

$$\text{rk}(\tilde{\mathbf{H}}\mathbf{P}\mathbf{X}) = \text{rk}(\mathbf{G}(\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y})) = \text{rk}(\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}).$$

In particular, setting $\mathbf{A} = \mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y} \in \mathbb{F}_q^{m \times t}$, we want that $\text{rk}(\mathbf{G}\mathbf{A}) = \text{rk}(\mathbf{A})$, given a full rank matrix $\mathbf{G}$. Consider $\mathbf{G}$ and $\mathbf{A}$ as linear maps, then

$$\text{rk}(\mathbf{G}\mathbf{A}) = \text{rk}(\mathbf{A}) - \dim(\text{RKer}(\mathbf{G}) \cap \text{Im}(\mathbf{A})).$$

Observe that, by definition, $\text{RKer}(\mathbf{G})$ is the space spanned by the columns of $\mathbf{Y}$. Then, the probability that the dimension of $\text{RKer}(\mathbf{G}) \cap \text{Im}(\mathbf{A})$ is non-zero is equal to the probability that two random subspaces of $\mathbb{F}_q^m$, of dimension t and $\text{rk}(\mathbf{A}) \leq t$, non-trivially intersect. Hence, since $t + \text{rk}(\mathbf{A}) \leq 2t < m$, this probability is given by $O(1/q^{m-t-\text{rk}(\mathbf{A})})$. $\square$

4 Algebraic Modelings

Our formulation makes gr-IPKP a *determinantal problem*, a family to which also MinRank and Rank Syndrome Decoding belong. Therefore, using the definition of gr-IPKP from Definition 6, we can take inspiration from algebraic attacks to these problems. Suppose that $n \geq m \geq t$, then the maximal rank attained by the $m \times t$ matrix $\mathbf{M}_{\mathbf{P}} = \mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}$ is t . Since by definition the matrix $\mathbf{M}_{\mathbf{P}}$ does not have maximum rank, it follows that every $t \times t$ minor of $\mathbf{M}_{\mathbf{P}}$ must vanish. In other words, every subset $I \subset \{1, \dots, m\}$ with $|I| = t$ leads to the following equation in $\mathbb{F}_q$

$$\det\left((\mathbf{M}_{\mathbf{P}})_{I,*}\right) = 0.$$

Furthermore, it is possible to obtain new equations by remembering that $\mathbf{P}$ is a permutation matrix. In order to do it, following a variant of [38, Sect. 3.2.3] we can add the equations which define a permutation matrix: $2n - 1$ independent linear equations $\sum_i x_{i,j} = 1$ and $\sum_i x_{j,i} = 1$ for every $j = 1, \dots, n$, together with

n^2 quadratic equations $x_{i,j}^2 - x_{i,j} = 0$ for every $i, j = 1, \dots, n$. In summary, we obtain the following polynomial system:

$$\begin{cases} \det \left((M_P)_{I,*} \right) = 0 & \text{for } I \in \binom{[m]}{t}, \\ \sum_{i=1}^n x_{i,j} = 1 & \text{for } j = 1, \dots, n, \\ \sum_{i=1}^n x_{j,i} = 1 & \text{for } j = 1, \dots, n, \\ x_{i,j}^2 - x_{i,j} = 0 & \text{for } i, j = 1, \dots, n. \end{cases} \quad (2)$$

To model equations $\det \left((M_P)_{I,*} \right) = 0$ for every $I \subset \{1, \dots, m\}$ with $|I| = t$ there are two possibilities: we can directly deal with the inhomogeneous case or, when possible, reduce it to the homogeneous case as reported in Proposition 10.

4.1 The Inhomogeneous Case

Recall that we are dealing with the condition $\text{rk}(M_P) = \text{rk}(HPX - Y) < t$, and that it is equivalent to saying that equation $\det \left((M_P)_{I,*} \right) = 0$ holds in $\mathbb{F}_q$ for each $I \in \binom{[m]}{t}$. We want to write the above equation in terms of P 's minors. First, we write

$$M_P = HPX - Y = (HPX \mid I_m) \begin{pmatrix} I_t \\ -Y \end{pmatrix}.$$

Then, since $\text{rk}(M_P) < t$, all its $\binom{m}{t}$ minors of dimension $t \times t$ must vanish

$$\det \left(\left((HPX \mid I_m) \begin{pmatrix} I_t \\ -Y \end{pmatrix} \right)_{I,*} \right) = 0 \quad \text{for } I \in \binom{[m]}{t}. \quad (3)$$

We can now use the Cauchy-Binet formula (Lemma 4) to rewrite Equation (3) as

$$\sum_{\substack{T \subset \{1, \dots, m+t\} \\ |T|=t}} \det \left((HPX \mid I_m)_{I,T} \right) \det \left(\begin{pmatrix} I_t \\ -Y \end{pmatrix}_{T,*} \right) = 0 \quad \text{for } I \in \binom{[m]}{t}.$$

Since the matrix Y is known, Equation (3) is a linear equation in which the unknowns are minors of the matrix $(HPX \mid I_m)$. In turn, these minors can be written as a linear combination of minors of P .

Moreover, we can see that some of the minors of the matrix $(HPX \mid I_m)$ are zero from its structure. In fact, let $T \in \binom{[t+m]}{t}$ and $T = T_1 \cup T_2$, where $T_1 \subset \{1, \dots, t\}$ and $T_2 \subset \{t+1, \dots, t+m\}$. Define $\tilde{T}_2 = \{i-t \mid i \in T_2\}$, then, the $(t+i)$ -th column of $(HPX \mid I_m)_{I, T_2}$, for all $i \in \tilde{T}_2 \setminus I$, contains all zeros:

$$I \begin{pmatrix} & & t+i \\ & 1 & 0 \\ HPX & & \ddots \\ & 0 & 1 \end{pmatrix}.$$

This fact can be seen in the following result, a slight modification of [3, Prop. 1], which shows that the minors of $\mathbf{HPX}$ can be written as a linear combination of some of the minors of $(\mathbf{HPX} \mid \mathbf{I}_m)$.

Lemma 11. *Let $\mathbf{A} \in \mathbb{F}_q^{m \times t}$, $T \in \binom{[t+m]}{t}$ and $I \in \binom{[m]}{t}$. Let $T = T_1 \cup T_2$ with $T_1 \subset \{1, \dots, m\}$ and $T_2 \subset \{m+1, \dots, m+t\}$. Consider $\tilde{T}_2 = \{i-t \mid i \in T_2\}$ then,*

$$\det \left((\mathbf{A} \mid \mathbf{I}_m)_{I,T} \right) = \begin{cases} 0 & \text{if } \tilde{T}_2 \not\subset I; \\ (-1)^{\sigma_I(T_2)} \det(-\mathbf{A}_{T_1, I \setminus T_2}) & \text{if } \tilde{T}_2 \subset I, \end{cases}$$

where $\sigma_I(T_2) = d(m+t) + (d-1)d/2 + \sum_{s \in \tilde{T}_2} \text{Pos}(s, I)$, and $\text{Pos}(s, I)$ is the position of s in the ordered set I and $d = |I| - |\tilde{T}_2|$.

We can then state the theorem concerning the algebraic modeling for gr-IPKP.

Theorem 12. *Let $(\mathbf{H}, \mathbf{X}, \mathbf{Y})$ be an gr-IPKP instance with parameters (q, n, m, t) . The condition $\text{rk}(\mathbf{HPX} - \mathbf{Y}) < t$ can be used to obtain a linear system in the minors of order up to t of $\mathbf{P}$ with $\binom{m}{t}$ equations and $\sum_{i=1}^t \binom{n}{i}^2$ unknowns.*

Proof. The equations of the system are the minors of order t of the matrix

$$\mathbf{HPX} - \mathbf{Y} = (\mathbf{HPX} \mid \mathbf{I}_m) \begin{pmatrix} \mathbf{I}_t \\ -\mathbf{Y} \end{pmatrix},$$

equated to 0. Let $I \in \binom{[m]}{t}$, $T \in \binom{[t+m]}{t}$, and let us write $T = T_1 \cup T_2$, with $T_1 \subseteq \{1 \dots t\}$ and $T_2 \subseteq \{t+1 \dots t+m\}$. For any choice of the subset I we have an equation, that is:

$$\begin{aligned} \det \left(\left((\mathbf{HPX} \mid \mathbf{I}_m) \begin{pmatrix} \mathbf{I}_t \\ -\mathbf{Y} \end{pmatrix} \right)_{I,*} \right) &= \\ \sum_{T \in \binom{[m+t]}{t}} \det \left((\mathbf{HPX} \mid \mathbf{I}_m)_{I,T} \right) \det \left(\begin{pmatrix} \mathbf{I}_t \\ -\mathbf{Y} \end{pmatrix}_{T,*} \right) &= \\ \sum_{T \in \binom{[m+t]}{t}} Y_T \det \left((\mathbf{HPX} \mid \mathbf{I}_m)_{I,T} \right), & \end{aligned} \quad (4)$$

where we first applied Lemma 4 and then we set Y_T as $\det \left(\begin{pmatrix} \mathbf{I}_t \\ -\mathbf{Y} \end{pmatrix}_{T,*} \right)$ for each T . We can now apply Lemma 11 to write Equation (4) as

$$\begin{aligned} & \sum_{\substack{T_1 \subset \{1, \dots, t\} \\ T_2 \subset \{t+1, \dots, t+m\} \\ |T_1| + |T_2| = t}} (-1)^{\sigma_I(T_2)} Y_{T_1 \cup T_2} \det \left((-\mathbf{H}\mathbf{P}\mathbf{X})_{I \setminus \tilde{T}_2, T_1} \right) = \\ & \sum_{\substack{T_1 \subset \{1, \dots, t\} \\ T_2 \subset \{t+1, \dots, t+m\} \\ |T_1| + |T_2| = t}} (-1)^{\sigma_I(T_2) + |T_1|} Y_{T_1 \cup T_2} \det \left((\mathbf{H}\mathbf{P}\mathbf{X})_{I \setminus \tilde{T}_2, T_1} \right), \end{aligned}$$

where, as usual $\tilde{T}_2 = \{i - t \mid i \in T_2\}$, and the integer $\sigma_I(T_2)$ is a function of I and T_2 defined in Lemma 11. Observe that the above equation has some terms defined only when T_2 is such that $\tilde{T}_2 \subset I$, otherwise the term is zero. Hence, we obtain

$$\sum_{\substack{T_1 \subset \{1, \dots, t\} \\ T_2 \subset \{t+1, \dots, t+m\} \\ \tilde{T}_2 \subset I, |T_1| + |T_2| = t}} (-1)^{\sigma_I(T_2) + |T_1|} Y_{T_1 \cup T_2} \det \left((\mathbf{H}\mathbf{P}\mathbf{X})_{I \setminus \tilde{T}_2, T_1} \right).$$

After applying the Cauchy-Binet formula twice (Lemma 4) and setting $X_{U, T_1} = \det(\mathbf{X}_{U, T_1})$ and $H_{I, T, V} = \det(\mathbf{H}_{I \setminus \tilde{T}_2, V})$, we obtain

$$\sum_{\substack{T_1 \subset \{1, \dots, t\} \\ T_2 \subset \{t+1, \dots, t+m\} \\ |T_1| + |T_2| = t}} (-1)^{\sigma_I(T_2) + |T_1|} Y_{T_1 \cup T_2} \sum_{U \in \binom{[n]}{|T_1|}} X_{U, T_1} \sum_{V \in \binom{[n]}{|T_1|}} H_{I, T_1 \cup T_2, V} \det(\mathbf{P}_{V, U}) = 0.$$

Therefore, there are $\binom{m}{t}$ equations, one for each choice of the set I . The unknowns are the minors $\det(\mathbf{P}_{V, U})$ and, in total, they are

$$\sum_{i=1}^t \binom{n}{i}^2$$

since i runs over all the possible cardinalities of the set T_1 and there are $\binom{n}{i}^2$ minors of $\mathbf{P}$ having order i . $\square$

To conclude the description of the modeling, we provide a compact representations of the equations in Theorem 12. Notice that, for $|V| = |U| = t_1$, $\det(\mathbf{P}_{V, U})$ is the (V, U) entry of the compound matrix $C_{t_1}(\mathbf{P})$, and its coefficient in the I -th equation is given by

$$M_{V, U}^{(I, t_1)} := \sum_{\substack{T_1 \subset \{1, \dots, t\} \\ |T_1| = t_1}} \sum_{\substack{T_2 \subset \{t+1, \dots, t+m\} \\ |T_2| = t - t_1}} (-1)^{\sigma_I(T_2) + |T_1|} Y_{T_1 \cup T_2} X_{U, T_1} H_{I, T_1 \cup T_2, V}.$$

We define $\mathbf{M}^{(I,t_1)}$ as the vector of length $\binom{n}{t_1}^2$ having entries indexed by $\binom{[n]}{t_1} \times \binom{[n]}{t_1}$ such that $(\mathbf{M}^{(I,t_1)})_{V,U} = M_{V,U}^{(I,t_1)}$. Hence, the system can be written with respect to t_1

$$\sum_{t_1=1}^t \mathbf{M}^{(I,t_1)} \text{vec}(C_{t_1}(\mathbf{P})) = (\mathbf{M}^{(I,1)} \mid \dots \mid \mathbf{M}^{(I,t)}) \begin{pmatrix} \text{vec}(C_1(\mathbf{P})) \\ \vdots \\ \text{vec}(C_t(\mathbf{P})) \end{pmatrix} = 0 \quad \text{for every } I,$$

where $\text{vec}(\mathbf{A})$ is the column vector obtained by taking the matrix $\mathbf{A}$ row by row. Let $N = \binom{m}{t}$ and let $I_1, \dots, I_N$ be all the possible elements of $\binom{[m]}{t}$, then the linear system having variables the entries of $C_1(\mathbf{P}), \dots, C_t(\mathbf{P})$ is given by

$$\begin{pmatrix} \mathbf{M}^{(I_1,1)} & \dots & \mathbf{M}^{(I_1,t)} \\ \vdots & \ddots & \vdots \\ \mathbf{M}^{(I_N,1)} & \dots & \mathbf{M}^{(I_N,t)} \end{pmatrix} \begin{pmatrix} \text{vec}(C_1(\mathbf{P})) \\ \vdots \\ \text{vec}(C_t(\mathbf{P})) \end{pmatrix} = \mathbf{0}.$$

4.2 The Homogeneous Case

From Proposition 10, starting from an gr-IPKP instance $(\mathbf{H}, \mathbf{X}, \mathbf{Y})$ with parameters (q, n, m, t) , we can obtain an gr-PKP instance $(\tilde{\mathbf{H}}, \mathbf{X})$ with parameters $(q, n, m - t, t)$. We show that, having an homogeneous instance, applying the same techniques from Section 4.1, we can get a linear system with less unknowns with respect to the one from Theorem 12.

Theorem 13. *Let $(\mathbf{H}, \mathbf{X})$ be an gr-PKP instance with parameters (q, n, m, t) . The condition $\text{rk}(\mathbf{H}\mathbf{P}\mathbf{X}) < t$ can be used to obtain a linear system with $\binom{m}{t}$ equations and $\binom{n}{t}^2$ unknowns.*

Proof. The condition $\text{rk}(\mathbf{H}\mathbf{P}\mathbf{X}) < t$ is equivalent to saying that

$$\det((\mathbf{H}\mathbf{P}\mathbf{X})_{I,*}) = 0$$

holds in $\mathbb{F}_q$ for each subset $I \in \binom{[m]}{t}$. Hence, using the Cauchy-Binet formula twice (Lemma 4), we obtain

$$\begin{aligned} \det((\mathbf{H}\mathbf{P}\mathbf{X})_{I,*}) &= \sum_{T \in \binom{[n]}{t}} \det((\mathbf{H}\mathbf{P})_{I,T}) \det(\mathbf{X}_{T,*}) = \\ &= \sum_{T \in \binom{[n]}{t}} \det(\mathbf{X}_{T,*}) \sum_{S \in \binom{[n]}{t}} \det(\mathbf{H}_{I,S}) \det(\mathbf{P}_{S,T}) = \\ &= \sum_{T, S \in \binom{[n]}{t}} \det(\mathbf{H}_{I,S}) \det(\mathbf{P}_{S,T}) \det(\mathbf{X}_{T,*}) = 0. \end{aligned}$$

The above linear system has $\binom{m}{t}$ equations and $\binom{n}{t}^2$ unknowns, which are the $t \times t$ minors of the permutation matrix $\mathbf{P}$. $\square$

As we did in Section 4.1, we write the system from Theorem 13 in a more compact way. Observe that the I -th equation can be written as

$$(C_t(\mathbf{H})_I \otimes C_t(\mathbf{X})^T) \text{vec}(C_t(\mathbf{P})) = 0,$$

where $C_t(\mathbf{H})_I$ is the row of the t -th compound matrix of $\mathbf{H}$. Now, let $N = \binom{m}{t}$ and let $I_1, \dots, I_N$ be all the possible subsets in $\binom{[m]}{t}$. Then, the whole linear system is given by

$$\begin{pmatrix} C_t(\mathbf{H})_{I_1} \otimes C_t(\mathbf{X})^T \\ \vdots \\ C_t(\mathbf{H})_{I_N} \otimes C_t(\mathbf{X})^T \end{pmatrix} \text{vec}(C_t(\mathbf{P})) = \left(C_t(\mathbf{H}) \otimes C_t(\mathbf{X})^T \right) \text{vec}(C_t(\mathbf{P})) = \mathbf{0}.$$

In short, we have

$$C_t(\mathbf{H}) C_t(\mathbf{P}) C_t(\mathbf{X}) = \mathbf{0}$$

as a mono-dimensional (signed-permutation) PKP instance with parameters $(q, \binom{n}{t}, \binom{m}{t}, 1)$. Observe that, as proven later in Proposition 15, when the field $\mathbb{F}_q$ has characteristic 2, $C_t(\mathbf{P})$ is a permutation matrix and we get a PKP instance.

4.3 Estimating the Practical Complexity

Now, we give some estimates on the practical complexity of solving systems of the form

$$\text{rk}((\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y})_{I,*}) < t, \quad (5)$$

where $\mathbf{P}$ is constrained to be a permutation matrix.

We first observe that an instance of the generalized relaxed Permuted Kernel Problem is an instance of the generalized MinRank problem [25, 16] where the solution space is intersected with the variety of permutation matrices. We recall here below its definition, since it will be used in the following.

Definition 14 (gMinRank). *Let m, n, k, r be positive integers and q be a prime power. The generalized MinRank Problem (gMinRank) asks on input $\mathbf{M}(x_1, \dots, x_k)$ in $\mathbb{F}_q[x_1, \dots, x_k]^{m \times n}$ to find $(\lambda_1, \dots, \lambda_k) \in \mathbb{F}_q^k$ such that*

$$\text{rk}(\mathbf{M}(\lambda_1, \dots, \lambda_k)) \leq r.$$

We will classify generalized MinRank instances as follows:

- a gMinRank instance is *under-defined* if $k > (n - r)(m - r)$;
- a gMinRank instance is *well-defined* if $k = (n - r)(m - r)$;
- a gMinRank instance is *over-determined* if $k < (n - r)(m - r)$.

In our case, to solve Equation (5), if $\mathfrak{I}_{\text{rk}}$ denotes the ideal associated to Equation (5) and $\mathfrak{I}_{\mathcal{S}_n}$ is the ideal associated to the permutation matrices constraints, we are interested in the ideal $\mathfrak{I} := \mathfrak{I}_{\text{rk}} + \mathfrak{I}_{\mathcal{S}_n}$ and in the solving degree of the set of polynomials given by the union of their generators. Since in our case the

polynomial system and the ideal $\mathfrak{J}$ have a very particular structure, especially for the equations arising from the permutation matrix constraints, we focus on the Castelnuovo-Mumford regularity. The Castelnuovo-Mumford regularity of the ideal $\mathfrak{J}$ generated by the set $\mathcal{F}$ is denoted with $\text{reg}(\mathfrak{J})$ and is a rather technical definition which can be found, for example, in [19]. It is known that it upper bounds the solving degree [17, Th. 10]:

$$d_{\text{sol}}(\mathcal{F}) \leq \text{reg}(\mathfrak{J}).$$

For this reason, since our ideal $\mathfrak{J}$ is the sum of $\mathfrak{J}_{\text{rk}}$ and $\mathfrak{J}_{\mathcal{S}_n}$ we estimate its Castelnuovo-Mumford regularity using the ones of its addends.

The ideal $\mathfrak{J}_{\text{rk}}$. We start analyzing the ideal $\mathfrak{J}_{\text{rk}}$, modeling the constraint on the rank of the matrix $\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}$ in the variables $\mathbf{P}_{i,j}$. We observe that it is exactly the ideal associated to an instance of the generalized MinRank problem.

Given an r-IPKP instance with parameters (q, n, m, t) , it is easy to see that $\mathfrak{J}_{\text{rk}}$ is the ideal associated to the generalized MinRank instance given by the matrix $\mathbf{H}\mathbf{P}\mathbf{X} - \mathbf{Y}$ in $\mathbb{F}_q[\mathbf{P}_{1,1}, \dots, \mathbf{P}_{n,n}]^{m \times t}$, with target degree $t - 1$. In particular, since in the parameters sets n is usually greater than m , we have

$$n^2 > (m - t + 1)(t - t + 1) = (m - t + 1),$$

implying an under-defined generalized MinRank instance. The solving degree of such systems has been extensively studied in [25, 16]. The Castelnuovo-Mumford regularity of the ideal $\mathfrak{J}_{\text{rk}}$ is computed in the proof of [16, Th. 6], showing that

$$\text{reg}(\mathfrak{J}_{\text{rk}}) = t.$$

The ideal $\mathfrak{J}_{\mathcal{S}_n}$. The ideal $\mathfrak{J}_{\mathcal{S}_n} \subset \mathcal{R} := \mathbb{F}_q[\mathbf{P}_{1,1}, \dots, \mathbf{P}_{n,n}]$ is generated by the polynomials

$$\begin{cases} \sum_{i=1}^n \mathbf{P}_{i,j} - 1 & \text{for } j = 1, \dots, n, \\ \sum_{j=1}^n \mathbf{P}_{i,j} - 1 & \text{for } i = 1, \dots, n, \\ \mathbf{P}_{i,j}^2 - \mathbf{P}_{i,j} & \text{for } i, j = 1, \dots, n. \end{cases}$$

It can be seen that it does not depend on the particular gr-IPKP instance, and hence we can assume that the polynomials are unrelated from those given in Equation (5). Notice that $\mathfrak{J}_{\mathcal{S}_n}$ is a zero-dimensional ideal, and it vanishes on $n!$ points corresponding to the possible permutation matrices of order n . For a zero-dimensional ideal I , the Hilbert function stabilizes at the value $|V(I)|$ corresponding to the cardinality of the variety. Therefore, the Castelnuovo-Mumford regularity of $\mathfrak{J}_{\mathcal{S}_n}$ is exactly the first degree d such that

$$\text{HF}_{\mathcal{R}/\mathfrak{J}_{\mathcal{S}_n}}(d) = |V(\mathfrak{J}_{\mathcal{S}_n})| = n!.$$

Now we prove that $d = n - 1$ by showing that it is possible to construct a basis for the quotient $\mathcal{R}/\mathfrak{J}_{\mathcal{S}_n}$ with $n!$ monomials of degree $n - 1$. Let us consider the

monomials

$$F_\sigma(\mathbf{P}_{1,1}, \dots, \mathbf{P}_{n,n}) = \prod_{i=1}^{n-1} P_{i,\sigma(i)}, \quad (6)$$

where $\sigma \in \mathcal{S}_n$ ranges among all permutations of the n indices. They are exactly $n!$ monomials of degree $n - 1$. Moreover, they distinguish all $n!$ permutation matrices. In fact, $F_\sigma(\mathbf{P}_\sigma) = 1$, for the permutation matrix $\mathbf{P}_\sigma$ having the 1's in position $(1, \sigma(1)), \dots, (n, \sigma(n))$, and $F_\sigma(\mathbf{P}) = 0$ for $\mathbf{P} \neq \mathbf{P}_\sigma$. Therefore, the functions F_σ are all linearly independent and span $\mathcal{R}/\mathcal{I}_{\mathcal{S}_n}$. It is not hard to see that $n - 2$ polynomials can not be sufficient to achieve that, as we do not have a unique permutation matrix by fixing $n - 2$ rows. In fact, for any fixed $n - 2$ rows, there exist at least two different permutations that the monomials of degree $n - 2$ can not distinguish. Therefore, the Castelnuovo-Mumford regularity of the permutation ideal is exactly

$$\text{reg}(\mathcal{I}_{\mathcal{S}_n}) = n - 1.$$

Estimating the solving degree of $\mathcal{I}$. Now that we have an estimation for both $\text{reg}(\mathcal{I}_{\text{rk}})$ and $\text{reg}(\mathcal{I}_{\mathcal{S}_n})$, we can deal with $\text{reg}(\mathcal{I}) = \text{reg}(\mathcal{I}_{\text{rk}} + \mathcal{I}_{\mathcal{S}_n})$.

The short exact sequence

$$0 \rightarrow \mathcal{I}_{\text{rk}} \cap \mathcal{I}_{\mathcal{S}_n} \rightarrow \mathcal{I}_{\text{rk}} \oplus \mathcal{I}_{\mathcal{S}_n} \rightarrow \mathcal{I}_{\text{rk}} + \mathcal{I}_{\mathcal{S}_n} \rightarrow 0$$

gives the inequality [21, Prop. 20.19]

$$\text{reg}(\mathcal{I}) = \text{reg}(\mathcal{I}_{\text{rk}} + \mathcal{I}_{\mathcal{S}_n}) \leq \max\{\text{reg}(\mathcal{I}_{\text{rk}} \oplus \mathcal{I}_{\mathcal{S}_n}), \text{reg}(\mathcal{I}_{\text{rk}} \cap \mathcal{I}_{\mathcal{S}_n}) - 1\}.$$

By definition, the Castelnuovo-Mumford regularity of a direct sum of ideals is the maximum between their regularities [42], namely

$$\text{reg}(\mathcal{I}_{\text{rk}} \oplus \mathcal{I}_{\mathcal{S}_n}) = \max\{\text{reg}(\mathcal{I}_{\text{rk}}), \text{reg}(\mathcal{I}_{\mathcal{S}_n})\}.$$

Concerning the ideal $\mathcal{I}_{\text{rk}} \cap \mathcal{I}_{\mathcal{S}_n}$, a rough upper bound in terms of the regularities of $\mathcal{I}_{\text{rk}}$ and $\mathcal{I}_{\mathcal{S}_n}$ is given by their sum. As noted in [20], in generic position this bound is believed to hold, and in the same work, to give counterexamples, ad-hoc constructions are needed. Since the varieties associated to $\mathcal{I}_{\text{rk}}$ and $\mathcal{I}_{\mathcal{S}_n}$ have just a point in common and since $\mathcal{I}_{\text{rk}}$ is random and instance-dependent (in contraposition to $\mathcal{I}_{\mathcal{S}_n}$ which depends only on n), we can assume that a bound on the Castelnuovo-Mumford regularity of the intersection is given by

$$\text{reg}(\mathcal{I}_{\text{rk}} \cap \mathcal{I}_{\mathcal{S}_n}) \leq \text{reg}(\mathcal{I}_{\text{rk}}) + \text{reg}(\mathcal{I}_{\mathcal{S}_n}).$$

Summing up, we have that

$$\text{reg}(\mathcal{I}) \leq \max\{\text{reg}(\mathcal{I}_{\text{rk}}), \text{reg}(\mathcal{I}_{\mathcal{S}_n}), \text{reg}(\mathcal{I}_{\text{rk}}) + \text{reg}(\mathcal{I}_{\mathcal{S}_n}) - 1\}$$

which implies, using the estimations from the previous subsections,

$$\text{d}_{\text{sol}}(\mathcal{F}) \leq \text{reg}(\mathcal{I}) \leq \max\{t, n - 1, t + n - 2\} = t + n - 2,$$

where $\mathcal{F}$ is the set of polynomials from Equation (2). Therefore, the number of binary operations to compute a Gröbner basis with F4/F5 for the ideal $\mathfrak{J}$ generated by the set of polynomials $\mathcal{F}$ can be estimated by

$$\mathcal{O}\left(\log_2(q)\binom{n^2+t+n-2}{t+n-2}^\omega\right).$$

4.4 A Toy Example

Consider the following gr-IPKP instance with $(q, n, m, t) = (7, 4, 3, 2)$.

$$\mathbf{H} = \begin{pmatrix} 5 & 1 & 2 & 3 \\ 5 & 4 & 0 & 5 \\ 4 & 4 & 3 & 3 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 4 & 2 \\ 6 & 1 \\ 5 & 0 \\ 2 & 5 \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} 0 & 2 \\ 5 & 2 \\ 4 & 5 \end{pmatrix}. \quad (7)$$

Then, we consider as unknowns the matrix $\mathbf{P} = (x_{i,j})$. The ideal $\mathfrak{J}_{S_4}$ is generated by polynomials

$$\begin{cases} x_{ij}^2 - x_{ij} & \text{for } i, j = 1, \dots, 4 \\ \sum_{i=1}^4 x_{ij} - 1 & \text{for } j = 1, \dots, 4 \\ \sum_{j=1}^4 x_{ij} - 1 & \text{for } i = 1, \dots, 4, \end{cases}$$

while the ideal $\mathfrak{J}_{rk}$ is generated by the $\binom{m}{t} = \binom{3}{2} = 3$ determinantal polynomials. For example, the one related to $I = \{1, 3\}$ is given by

$$\begin{aligned} & 2x_{1,2}x_{2,1} + 6x_{1,3}x_{2,1} + 3x_{1,4}x_{2,1} + 5x_{1,1}x_{2,2} + 3x_{1,3}x_{2,2} + x_{1,1}x_{2,3} + 4x_{1,2}x_{2,3} \\ & + 6x_{1,4}x_{2,3} + 4x_{1,1}x_{2,4} + x_{1,3}x_{2,4} + 2x_{2,2}x_{3,1} + 6x_{2,3}x_{3,1} + 3x_{2,4}x_{3,1} + 5x_{2,1}x_{3,2} \\ & + 3x_{2,3}x_{3,2} + x_{2,1}x_{3,3} + 4x_{2,2}x_{3,3} + 6x_{2,4}x_{3,3} + 4x_{2,1}x_{3,4} + x_{2,3}x_{3,4} + 3x_{1,2}x_{4,1} \\ & + 2x_{1,3}x_{4,1} + x_{1,4}x_{4,1} + 5x_{2,2}x_{4,1} + x_{2,3}x_{4,1} + 4x_{2,4}x_{4,1} + 4x_{3,2}x_{4,1} + 5x_{3,3}x_{4,1} \\ & + 6x_{3,4}x_{4,1} + 4x_{1,1}x_{4,2} + x_{1,3}x_{4,2} + 2x_{2,1}x_{4,2} + 4x_{2,3}x_{4,2} + 3x_{2,4}x_{4,2} + 6x_{3,3}x_{4,2} \\ & + 5x_{1,1}x_{4,3} + 6x_{1,2}x_{4,3} + 2x_{1,4}x_{4,3} + 6x_{2,1}x_{4,3} + 3x_{2,2}x_{4,3} + x_{2,4}x_{4,3} + 2x_{3,1}x_{4,3} \\ & + 6x_{1,1}x_{4,4} + 5x_{1,3}x_{4,4} + 3x_{2,1}x_{4,4} + 6x_{2,3}x_{4,4} + x_{3,1}x_{4,4} + 2x_{3,3}x_{4,4} + 2x_{1,2} \\ & + 6x_{1,3} + 3x_{1,4} + 6x_{2,1} + x_{2,2} + x_{2,3} + 5x_{2,4} + 5x_{3,2} + x_{3,3} \\ & + 4x_{3,4} + 2x_{4,1} + 4x_{4,3} + 6. \end{aligned}$$

One can observe how the degree of such polynomials is $t = 2$ and, since $(\mathbf{H}, \mathbf{X}, \mathbf{Y})$ is an inhomogeneous instance, monomials of lower degrees appear too, concerning the minors of $\mathbf{P}$ of order 2 and 1. The computation of the Gröbner basis of $\mathfrak{J}_{rk} + \mathfrak{J}_{S_n}$ gives the following result

$$\{x_{1,1} + 6, x_{1,2}, x_{1,3}, x_{1,4}, x_{2,1}, x_{2,2}, x_{2,3} + 6, x_{2,4}, x_{3,1}, x_{3,2} + 6, x_{3,3}, x_{3,4}, x_{4,1}, x_{4,2}, x_{4,3}, x_{4,4} + 6\}.$$

One can see that the unique point in the related variety is given by the matrix

$$\tilde{\mathbf{P}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

and that the matrix

$$H\tilde{P}X - Y = \begin{pmatrix} 1 & 4 \\ 3 & 5 \\ 0 & 0 \end{pmatrix}$$

has rank less than $t = 2$, namely 1.

5 On the Minors of a Permutation Matrix

The modelings presented in Section 4 have a number of equations which is less than the number of unknowns, since in practice we always have $n \geq m$. However, it is possible to add more (non-linear) equations by looking more carefully at the structure of the minors of a permutation matrix.

It is known that every minor of a permutation matrix $\mathbf{P}$ is equal to 1, -1 , or 0. Hence, we can show that its t -th compound matrix (Definition 5) $C_t(\mathbf{P})$ has a special structure.

Proposition 15. *Let n, t be positive integers such that $t \leq n$ and let $\mathbf{P}$ be an $n \times n$ permutation matrix. The t -th compound matrix $C_t(\mathbf{P})$ is a signed permutation matrix.*

Proof. We want to prove that the matrix $C_t(\mathbf{P})$ has exactly one non-zero entry in each row and column, and that this non-zero entry is either 1 or -1 .

First of all, notice that the determinant of any square submatrix of a permutation matrix is either 0, 1 or -1 , hence $C_t(\mathbf{P}) \in \{-1, 0, 1\}^{\binom{n}{t} \times \binom{n}{t}}$. Furthermore, if one minor is non-zero, the considered submatrix of $\mathbf{P}$ is a permutation matrix itself.

Now, we prove that for a fixed $I \in \binom{[n]}{t}$, there exists a unique $J \in \binom{[n]}{t}$ such that $\mathbf{P}_{I,J}$ has a non-zero determinant. This is due to the permutation structure of $\mathbf{P}$. The existence of such a J is easy to observe: we can choose the columns indexes that contain the ones on the rows indexed by I . For uniqueness, suppose that there exist two J of this type, namely J_1 and J_2 , then the two matrices $\mathbf{P}_{I,J_1}$ and $\mathbf{P}_{I,J_2}$ are both permutation matrices, therefore there would exist at least one row indexed by I with two ones. This contradicts the fact that $\mathbf{P}$ is a permutation matrix. If we apply the same argument to the columns, we can conclude that $C_t(\mathbf{P})$ is a signed permutation matrix. $\square$

Thanks to this fact, when we consider the algebraic modelings of Section 4, we can add the equations of the signed permutation matrix $C_t(\mathbf{P})$: each entry is in $\{0, 1, -1\}$ leading to equations of the form $\det(\mathbf{P}_{I,J})^3 = \det(\mathbf{P}_{I,J})$, and the sums over the rows and the columns is ± 1 , i.e., $\sum_I \det(\mathbf{P}_{I,J})^2 = 1$ and $\sum_J \det(\mathbf{P}_{I,J})^2 = 1$.

At this point, one can wonder how to recover $\mathbf{P}$ from its compound matrix $C_t(\mathbf{P})$. In the modeling of Section 4.1, $C_1(\mathbf{P}) = \mathbf{P}$ appears in the equations, so we directly find the secret permutation matrix. However, for $t \neq 1$, from the modeling in Section 4.2 we only obtain $C_t(\mathbf{P})$. Assume that $t \neq n$ since in

that case $C_n(\mathbf{P}) = \det(\mathbf{P})$ and we cannot infer nothing about $\mathbf{P}$ beyond its determinant. However, the case $t \geq n$ leads to an easy instance of r-IPKP, as shown in [9]. From Proposition 15, we know that $C_t(\mathbf{P})$ is a signed permutation matrix, and hence, for each $I \in \binom{[n]}{t}$ there exists a unique $J_I \in \binom{[n]}{t}$ such that $(C_t(\mathbf{P}))_{I, J_I} \neq 0$. This means that $\mathbf{P}_{I, J_I}$ is a permutation matrix and the permutation $\pi_{\mathbf{P}}$ associated to $\mathbf{P}$ sends I to J_I , i.e. $\pi_{\mathbf{P}}(I) = J_I$. Hence, one can compute the image of $i \in [n]$ as

$$\pi_{\mathbf{P}}(i) = \bigcap_{\substack{I \in \binom{[n]}{t} \\ i \in I}} J_I = \bigcap_{\substack{I \in \binom{[n]}{t} \\ i \in I}} \pi_{\mathbf{P}}(I).$$

This procedure has a complexity of $O\left(\binom{n}{t}^2\right)$.

5.1 On Linearization

Under some hypothesis, a typical way to solve a polynomial system where the number of unknowns is larger than the number of equations consists in *linearizing* said system. In fact, modelings from Theorem 12 and Theorem 13 can be seen both as linear systems in the entries of $C_t(\mathbf{P})$ and as polynomial systems of degree t in the entries of $\mathbf{P}$. The former is the linearized system of the latter. We now show how to derive more equations at the cost of increasing the linear variables.

For simplicity, let us consider a homogeneous instance of r-PKP. The inhomogeneous one will be discussed at the end of this subsection. From Section 4.2, we have a linear system in the minors of order t of the permutation $\mathbf{P}$. We can use the *Leibnitz formula* to see the minors, i.e., the entries of $C_t(\mathbf{P})$, as homogeneous polynomials of degree t with $t!$ monomials in the entries of $\mathbf{P}$. Observe that different minors have a disjoint set of monomials. Hence, we have a total of $\binom{n}{t}^2 t!$ distinct monomials in our degree t equations. For example, consider $n = 4$ and $t = 2$, then

$$(C_2(\mathbf{P}))_{\{1,2\},\{1,3\}} = P_{11}P_{23} - P_{21}P_{13} = \pm 1.$$

Since $\mathbf{P}$ is a permutation matrix, we have that if the minor of order t is non-vanishing, then just one of its monomials is non-zero (namely, the one with all the variables equal to 1). Conversely, if the minor is zero, then each of its monomials is zero. The number of non-vanishing minors is equal to $\binom{n}{t}$, which is also the number of non-vanishing monomials. The number of vanishing monomials is then $\binom{n}{t}^2 t! - \binom{n}{t}$.

Linearizing, we obtain a homogeneous linear system with $\binom{n}{t}$ equations and $\binom{n}{t}^2 t!$ variables. One may wonder what is the benefit of using $\binom{n}{t}^2 t!$ variables instead of the $\binom{n}{t}^2$ ones from the previous modeling. Recall that a minor can only be in $\{-1, 0, 1\}$, and that only one minor does not vanish in each row and column of the matrix $C_t(\mathbf{P})$. This means that the sums over each column and row is equal to the value of the non-vanishing minor, i.e. it is in $\{-1, 0, 1\}$. Using

the monomials as variables, since they can only be in $\{0, 1\}$, the sum over the rows and columns of the monomials appearing in the minors must be 1 and not ± 1 . Hence, the previous *quadratic* equations of the form $\sum_I m_{I,J}^2 = 1$ and $\sum_J m_{I,J}^2 = 1$ become *linear* equations instead. Using the above example, for $n = 4$ and $t = 2$, then, taking $I = \{1, 2\}$, previously we had

$$\sum_{J \in \binom{[4]}{2}} (C_2(\mathbf{P}))_{\{1,2\},J} = \pm 1 \implies \left(\sum_{J \in \binom{[4]}{2}} (C_2(\mathbf{P}))_{\{1,2\},J} \right)^2 = 1.$$

Now, considering the monomials in $(C_2(\mathbf{P}))_{\{1,2\},J}$ as variables, we have the linear equation

$$\underbrace{P_{11}P_{22} + P_{21}P_{12}}_{(C_2(\mathbf{P}))_{\{1,2\},\{1,2\}}} + \underbrace{P_{11}P_{23} + P_{21}P_{13}}_{(C_2(\mathbf{P}))_{\{1,2\},\{1,3\}}} + \underbrace{P_{11}P_{24} + P_{21}P_{14}}_{(C_2(\mathbf{P}))_{\{1,2\},\{1,4\}}} + \underbrace{P_{12}P_{23} + P_{22}P_{13}}_{(C_2(\mathbf{P}))_{\{1,2\},\{2,3\}}} + \underbrace{P_{12}P_{24} + P_{22}P_{14}}_{(C_2(\mathbf{P}))_{\{1,2\},\{2,4\}}} + \underbrace{P_{13}P_{24} + P_{23}P_{14}}_{(C_2(\mathbf{P}))_{\{1,2\},\{3,4\}}} = 1,$$

where it is highlighted from which minor each monomial comes from.

Summing up, we obtain a linear system with $\binom{m}{t} + 2\binom{n}{t} - 1$ equations and $\binom{n}{t}^2 t!$ variables, of which we know that the majority of them vanish.

The inhomogeneous one can be adapted too. In this case, we get a linear system with $\binom{m}{t} + \sum_{i=1}^t (2\binom{n}{i} - 1)$ equations and $\sum_{i=1}^t \binom{n}{i}^2 i!$ variables.

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